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That's all folks

Terraform

ADVANCED

For the multi-cloud edition of this guide, click here

Welcome

Managing large numbers of settings in a webUI can be a total bummer. It'd be much nicer if we could describe our ZeroTier networks and membership settings as code. That would let us keep them in version control, and integrate them into our software delivery pipelines.

Now we can!

Terraform allows us to talk to the ZeroTier Central API and describe our network infrastructure, as code. This tutorial will walk you though how to get started.

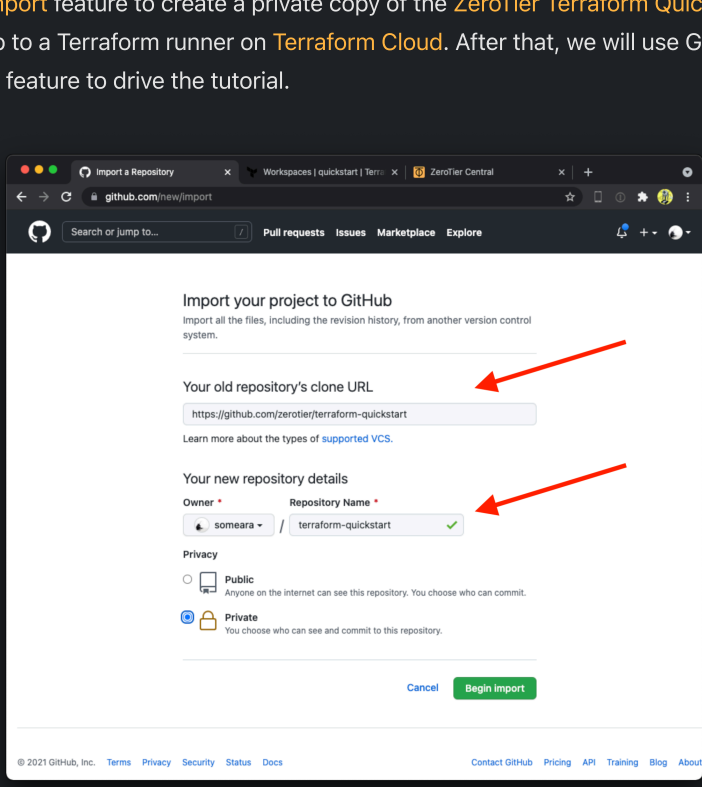
To follow along step by step, you will need:

- A Github account,
- A ZeroTier Central account,
- A Terraform Cloud account.

It should take you about 10 minutes to through this tutorial. It will be done *in browser* without touching the command line at all.

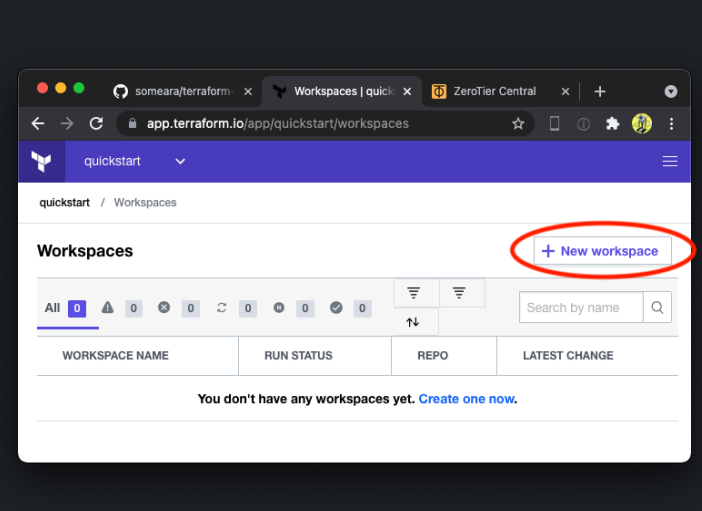
Import the Quickstart repo

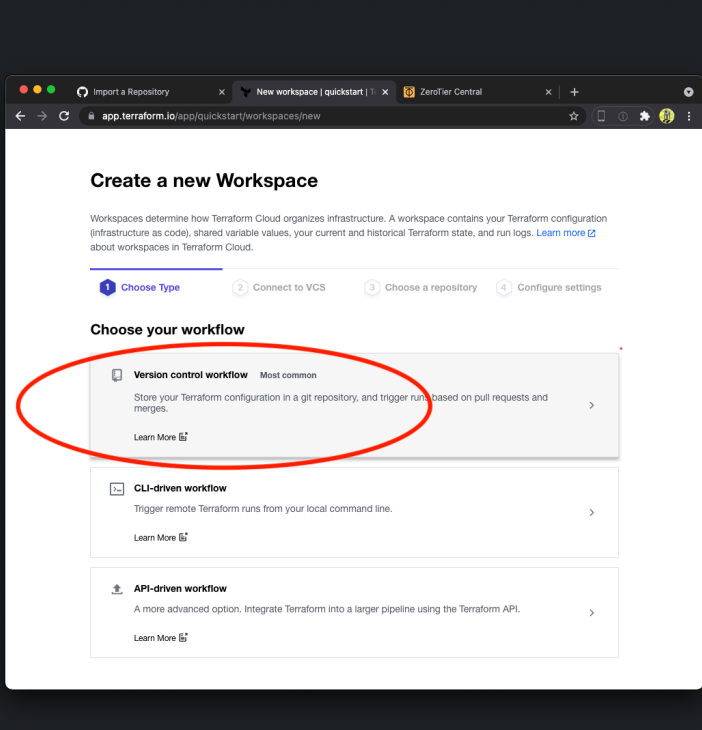
Use Github's Import feature to create a private copy of the ZeroTier Terraform Quickstart repo. We will hook this up to a Terraform runner on Terraform Cloud. After that, we will use Github's in-browser editing feature to drive the tutorial.

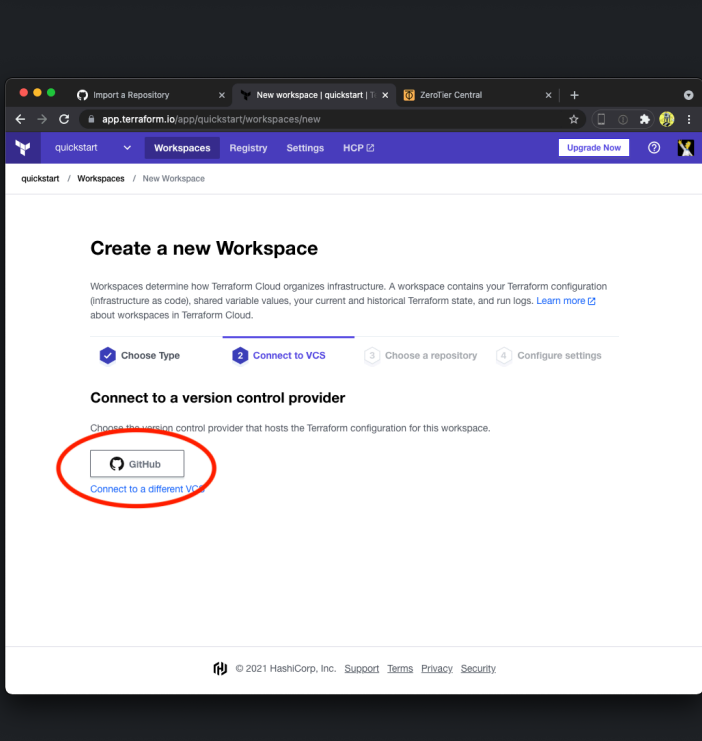


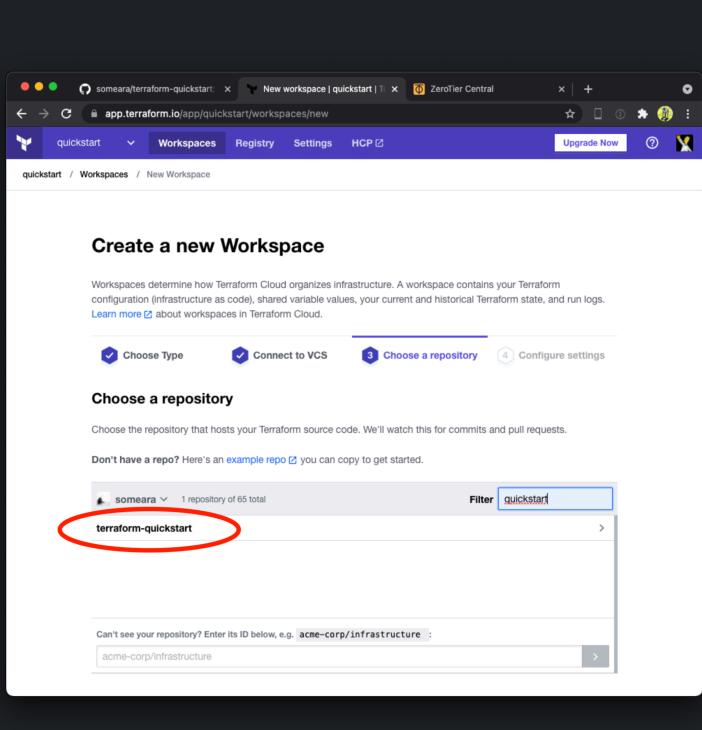
Create a Terraform workspace

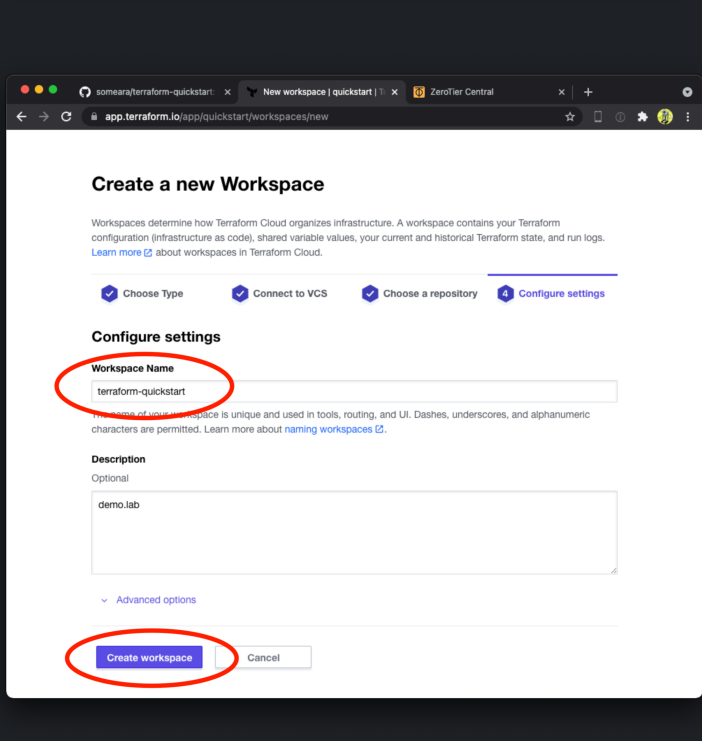
Next, we create a Terraform workspace and attach it to our private Github repository. Be sure to select ***version control workflow***, select the correct Github account, (we want the private copy, not the original), and give it a unique name.





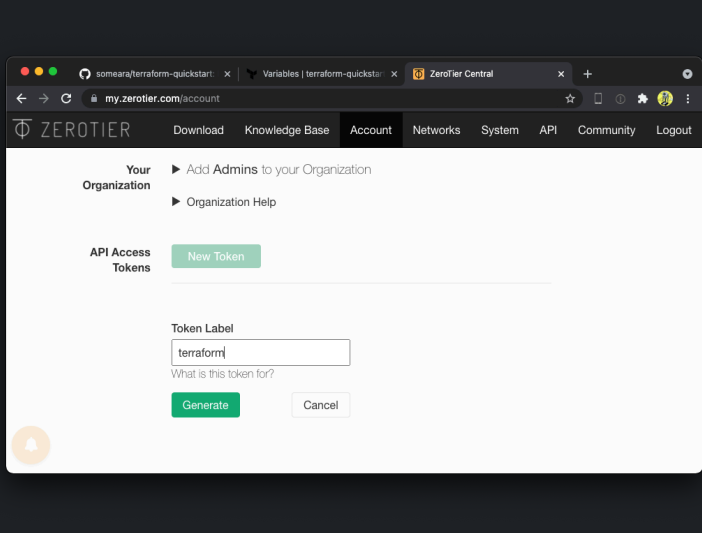






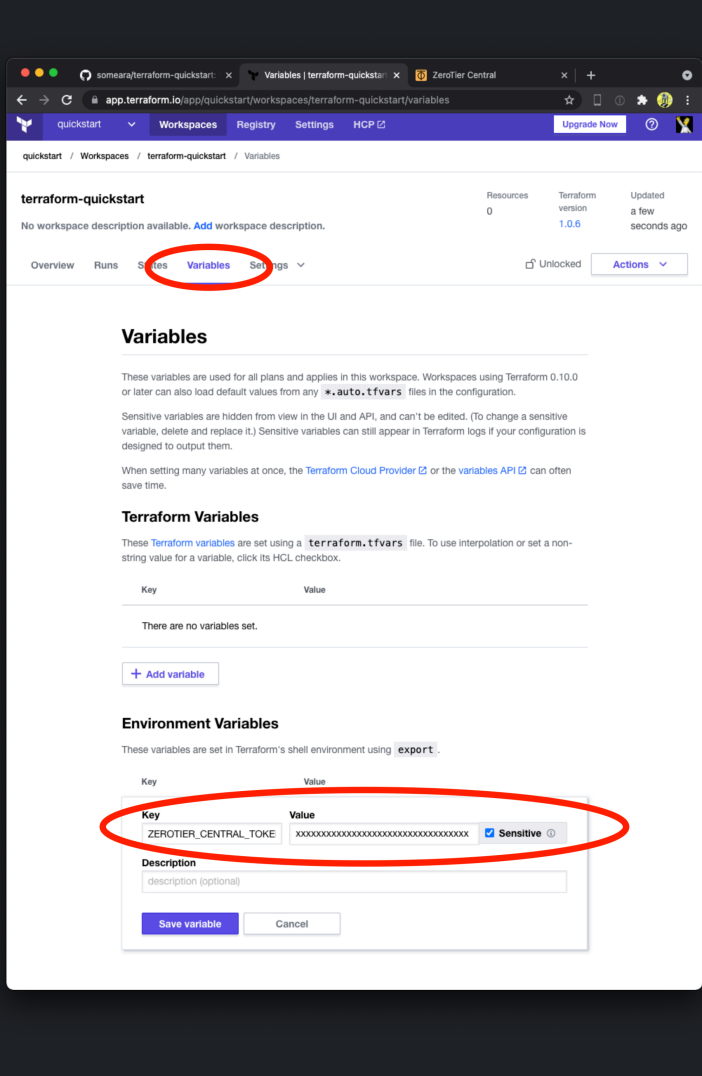
Create a Central API Token

Next, we create an API token that Terraform will use to drive the ZeroTier Central API. Navigate to `Account` -> `API Access Tokens`.



Add token as a Workspace Environment Variable

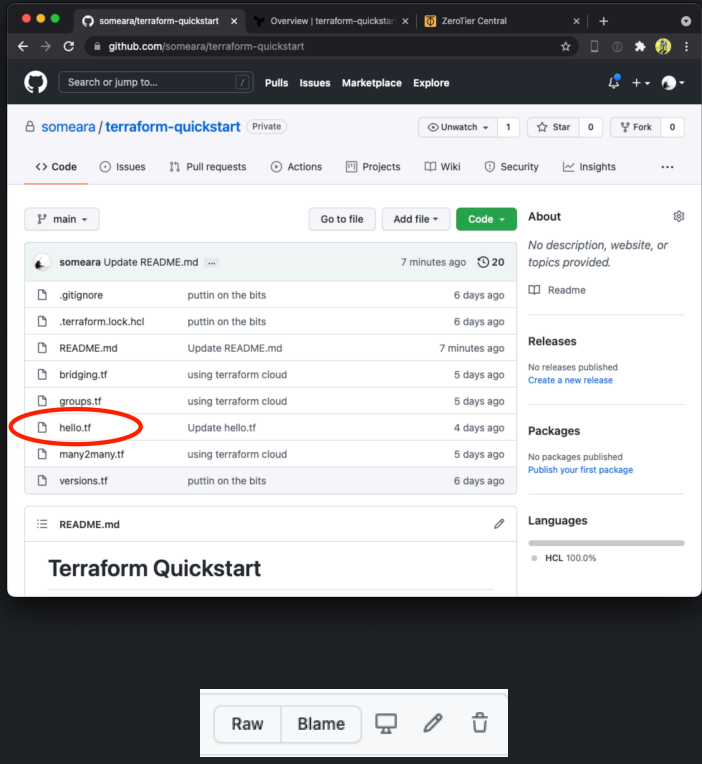
Add the token as an Environment Variable to our workspace. This will let the ZeroTier Terraform Provider authenticate to the API. The variable must be named `ZEROTIER_CENTRAL_TOKEN`. Be sure to check the `Sensitive` box.

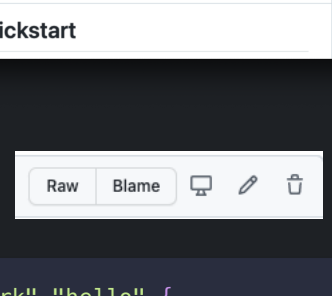


Hello World

This example is probably the simplest thing you can do with ZeroTier. It creates a single network, then joins two members. The `member_id` settings in the repository are made up, which is good enough to demonstrate how to drive the API with Terraform. Feel free to replace them with real Node IDs of any devices you may wish to join to the networks.

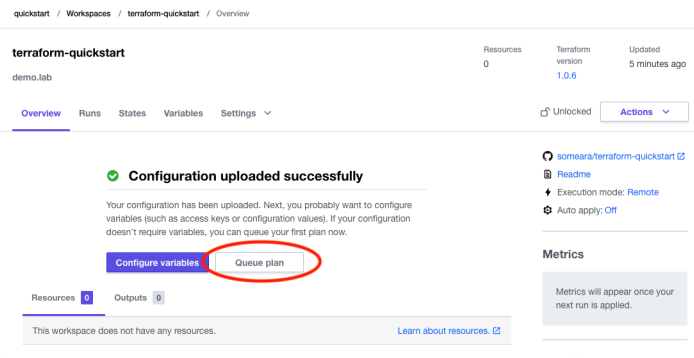
In your Github repo, click on `hello.tf`. There will be a little "edit" icon around the section with the code. Uncomment the Terraform resources and click the green "commit changes" button.

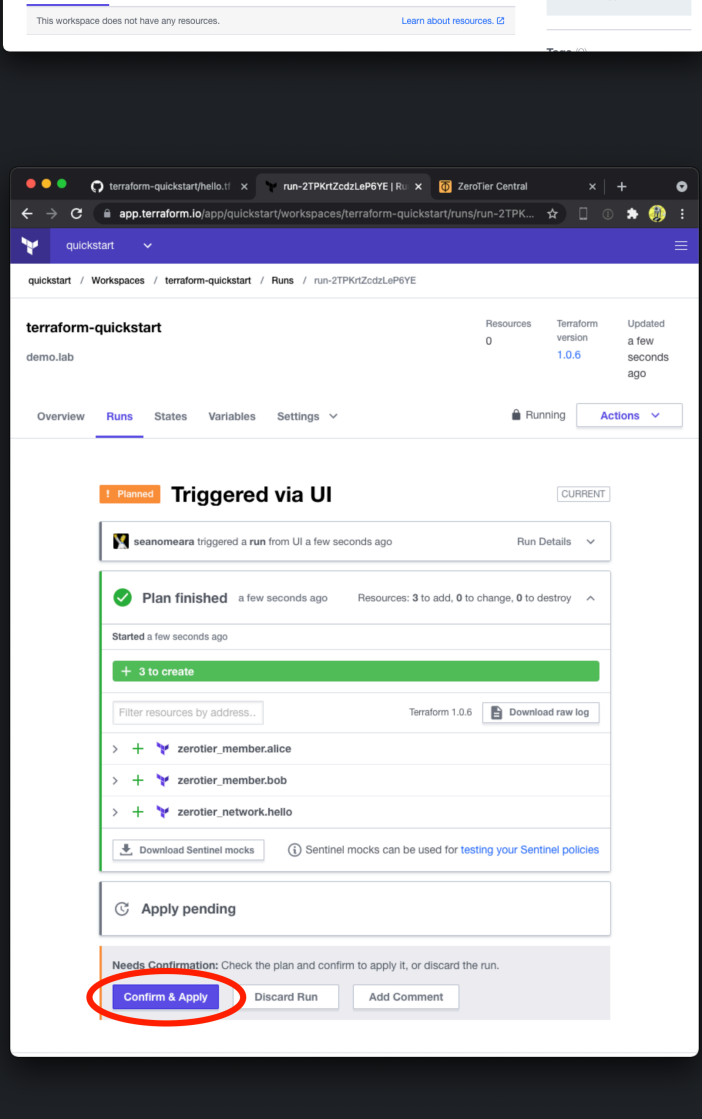




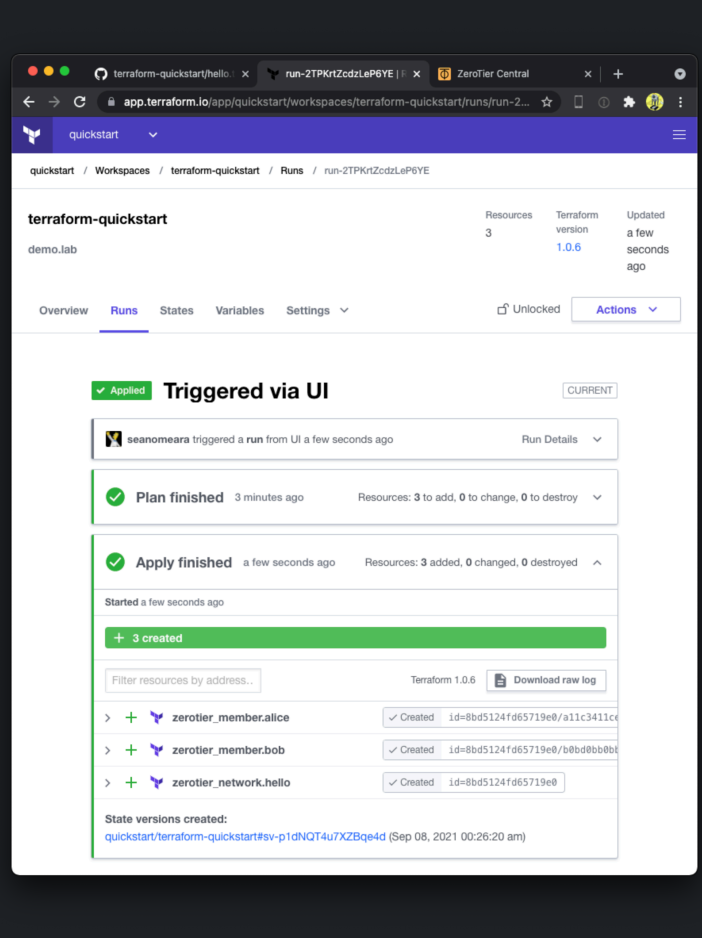
```
resource "zerotier_network" "hello" {
  name        = "hello"
  description = "Hello World"
  assignment_pool {
    start = "192.168.42.1"
    end   = "192.168.42.254"
  }
  route {
    target = "192.168.42.0/24"
  }
}

resource "zerotier_member" "alice" {
  name        = "alice"
  member_id   = "a11c3411ce"
  description = "Alice's laptop"
  network_id  = zerotier_network.hello.id
}

resource "zerotier_member" "bob" {
  name        = "bob"
  member_id   = "b0bd0bb0bb"
  description = "Bob's laptop"
  network_id  = zerotier_network.hello.id
}
```

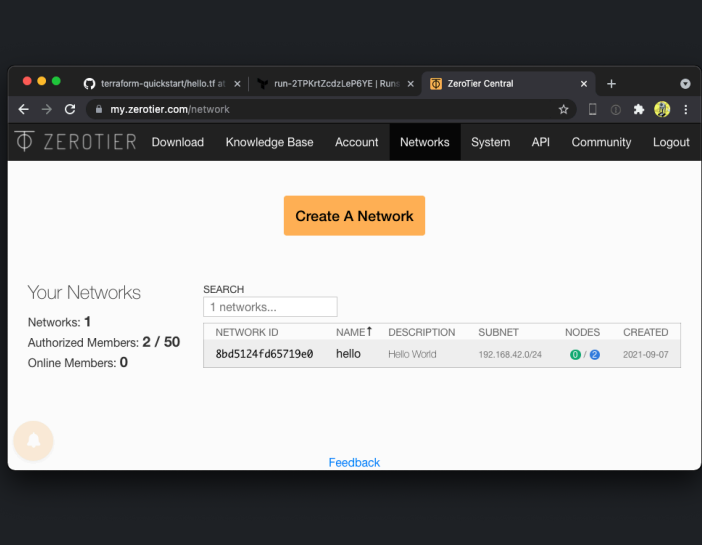
Queue a plan then "Confirm and Apply".





After Terraform applies the plan, check out the ZeroTier Cental webUI to confirm it was created.





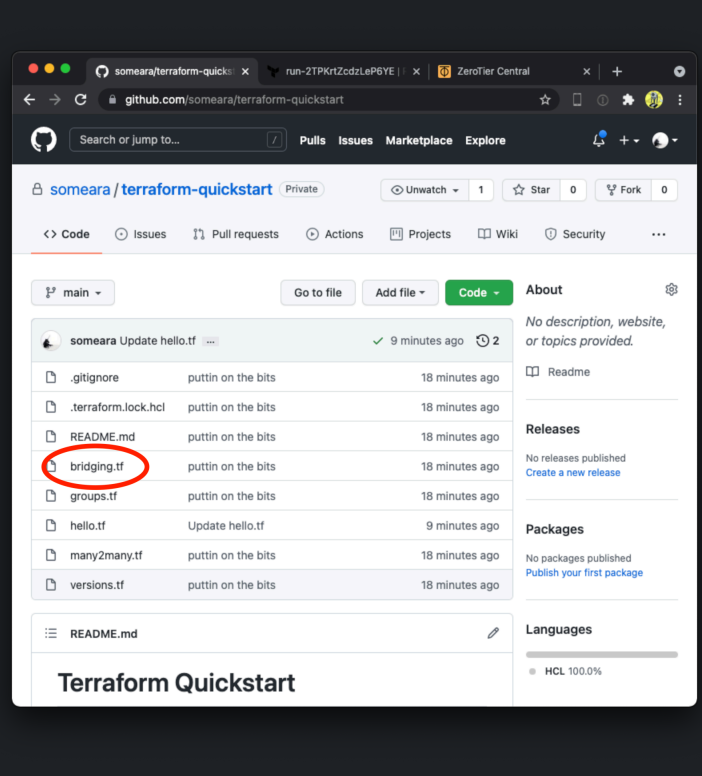
Bridging Networks

The next example manipulates the `allow_ethernet_bridging` settings on the Member objects. When running on machines with multiple physical Ethernet interfaces, ZeroTier can be configured to pass layer2 traffic such as ARP, NDP, multicast, mDNS, etc.

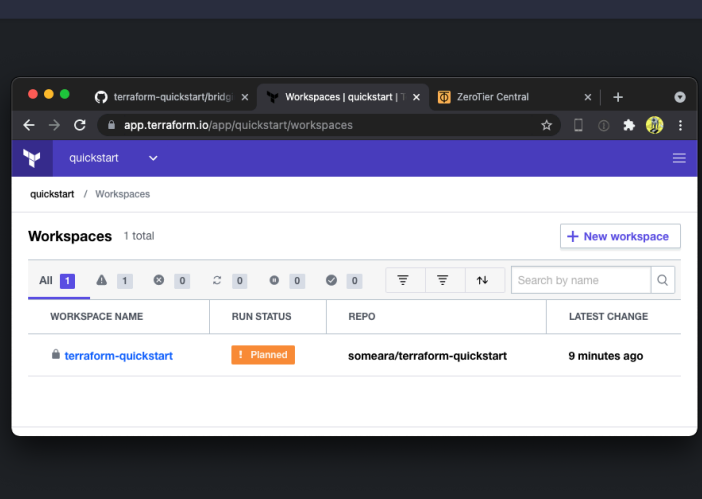
To make this work, you'll need to go into your router's OS and configure a bridge between a physical interface and the ZeroTier interface.

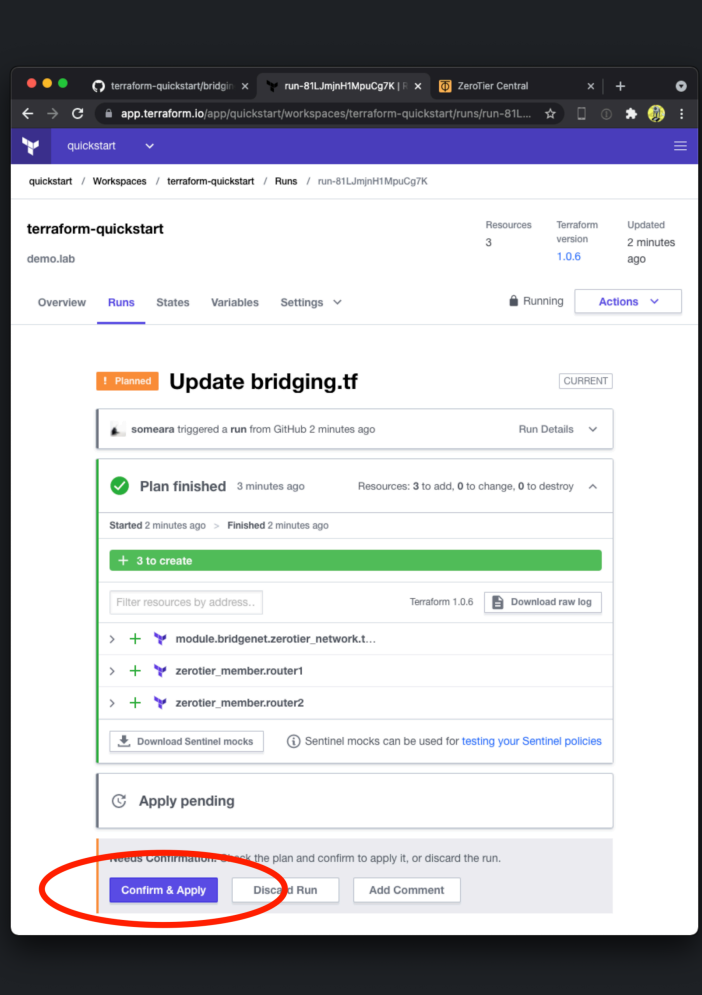
The ZeroTier Network Terraform module provides a slightly nicer interface, letting us use CIDRs for our subnets.

Repeat the steps from "Hello World" with `bridging.tf`

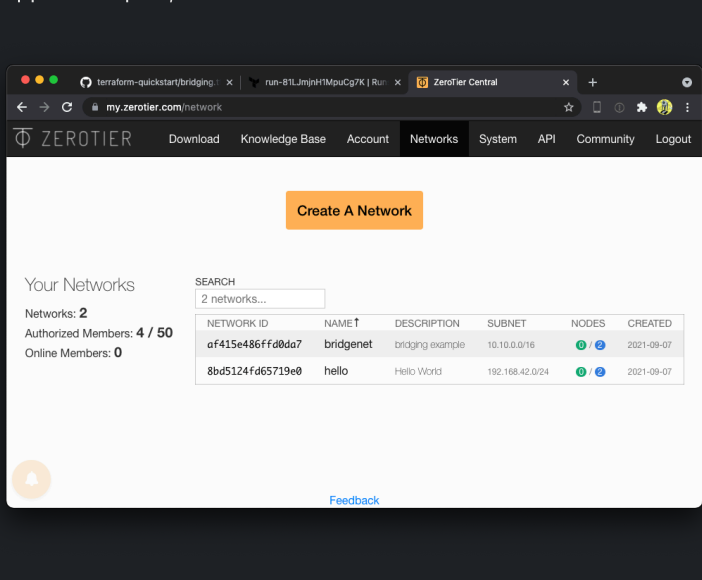


```
module "bridgenet" {
  source      = "zerotier/network/zerotier"
  version     = "1.0.0"
  name        = "bridgenet"
  description = "bridging example"
  subnets     = ["10.10.0.0/16"]
  flow_rules  = "accept;"
}

resource "zerotier_member" "router1" {
  name                    = "router1"
  member_id               = "71c71c71c1"
  description             = "Alice's router"
  ip_assignments          = ["10.10.1.1"]
  allow_ethernet_bridging = true
  network_id              = module.bridgenet.id
}

resource "zerotier_member" "router2" {
  name                    = "router2"
  member_id               = "71c71c71c2"
  description             = "Bob's router"
  ip_assignments          = ["10.10.2.1"]
  allow_ethernet_bridging = true
  network_id              = module.bridgenet.id
}
```





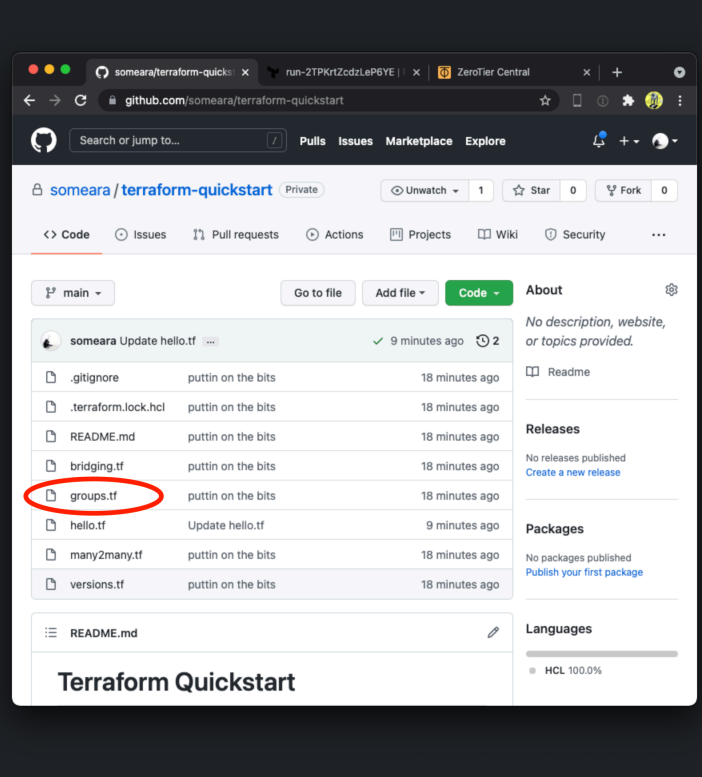
After Terraform applies the plan, check out the ZeroTier Cental webUI to confirm it was created.



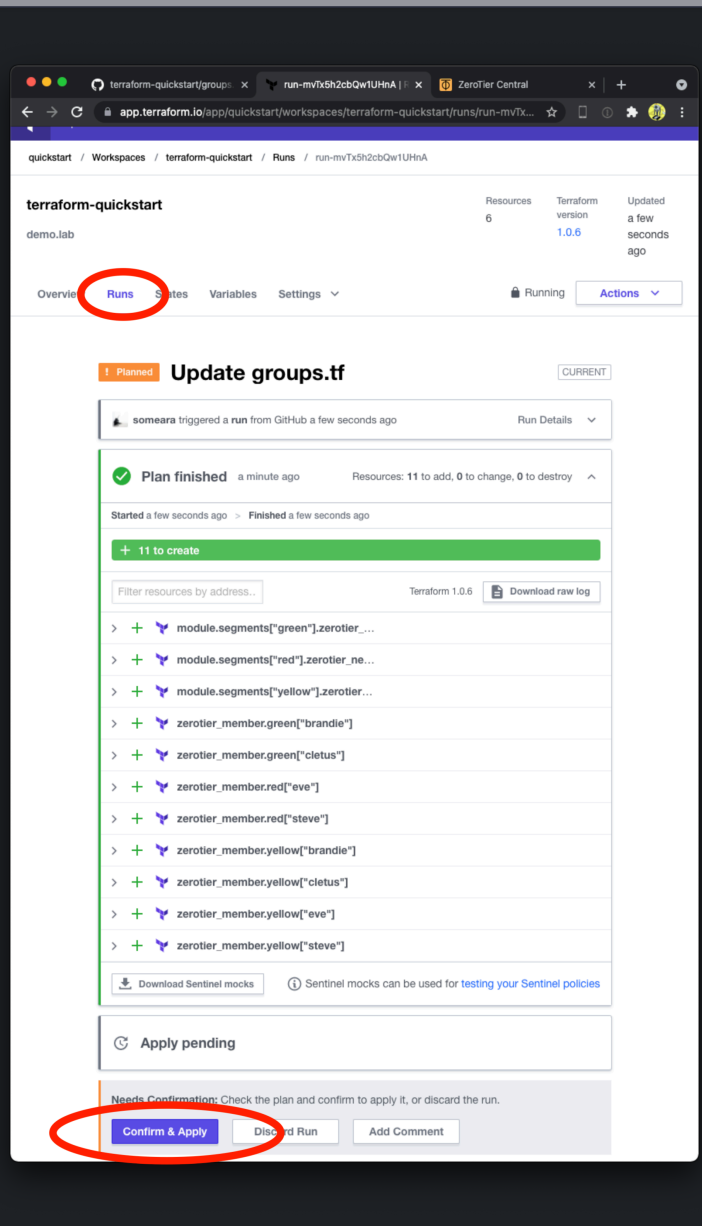
Network Segmentation

The next example creates the networks, `red`, `green`, and `yellow`. We define two groups. The red team gets access to the `red` network, and the green team gets access to the `green` network. Red and green make `yellow`.

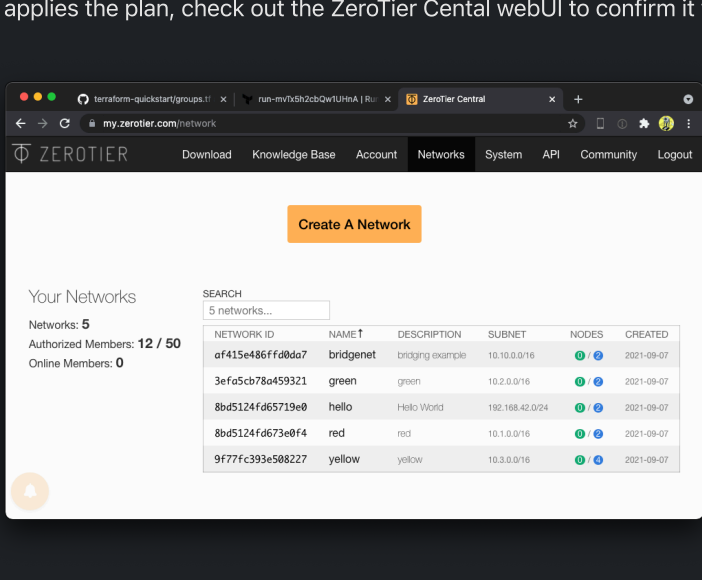
Repeat the steps from "Hello World" with `groups.tf`



```
variable "segments" {
  default = {
    red = {
      description = "red"
      subnets     = ["10.1.0.0/16"]
      flow_rules  = "accept;"
    }
    green = {
      description = "green"
      subnets     = ["10.2.0.0/16"]
      flow_rules  = "accept;"
    }
    yellow = {
      description = "yellow"
      subnets     = ["10.3.0.0/16"]
      flow_rules  = "accept;"
    }
  }
}

variable "members" {
  default = {
    red = {
      eve = {
        description = "Eve's Laptop"
        member_id   = "34b34b34b3"
      }
      steve = {
        description = "Steve's Laptop"
        member_id   = "573b3573b3"
      }
    }
    green = {
      cletus = {
        description = "Cletus' Laptop"
        member_id   = "c133715b0b"
      }
      brandie = {
        description = "Brandie's Laptop"
        member_id   = "b33fb33fff"
      }
    }
  }
}

module "segments" {
  for_each    = var.segments
  source      = "zerotier/network/zerotier"
  version     = "1.0.0"
  name        = each.key
  description = each.value.description
  subnets     = each.value.subnets
  flow_rules  = each.value.flow_rules
}

resource "zerotier_member" "red" {
  for_each   = { for k, v in var.members.red : (k) => v }
  name       = each.key
  member_id  = each.value.member_id
  network_id = module.segments["red"].id
}

resource "zerotier_member" "green" {
  for_each   = { for k, v in var.members.green : (k) => v }
  name       = each.key
  member_id  = each.value.member_id
  network_id = module.segments["green"].id
}

resource "zerotier_member" "yellow" {
  for_each   = { for k, v in merge(var.members.red, var.members.green) : (k
  name       = each.key
  member_id  = each.value.member_id
  network_id = module.segments["yellow"].id
}
```



After Terraform applies the plan, check out the ZeroTier Cental webUI to confirm it was created.

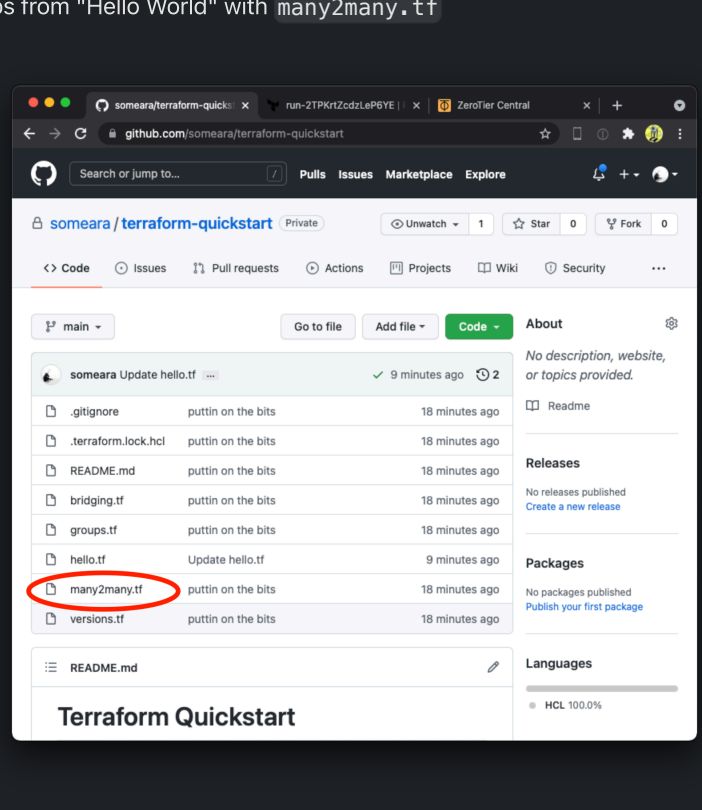


Many to Many

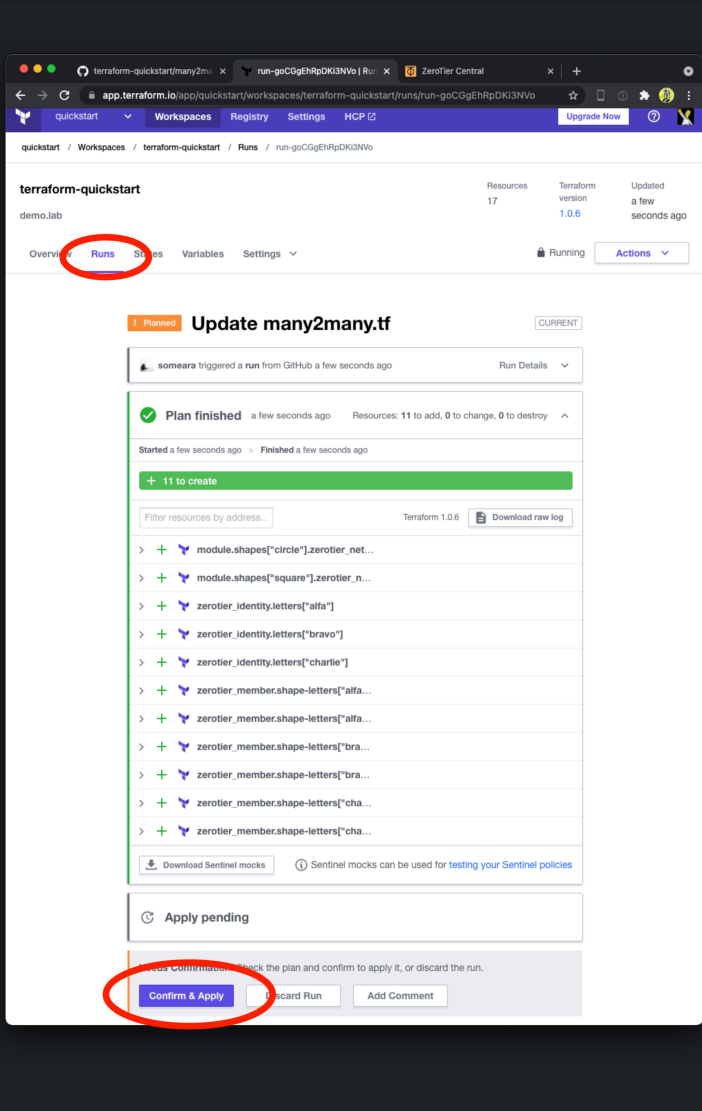
In the last example, show how to assign many members to many networks. This example used the Terraform setproduct function to find all possible combinations of two sets.

The `zerotier_identity` resource is a distant cousin of the Terraform tls_private_key resource. This resource would normally be used to inject secrets into cloud instances via `cloudinit`. We encourage you to use the Terraform Cloud to keep your Terraform state safe.

Repeat the steps from "Hello World" with `many2many.tf`



```
variable "letters" {
  default = ["alfa", "bravo", "charlie"]
}

variable "shapes" {
  default = ["circle", "square", "diamond" ]
}

resource "zerotier_identity" "letters" {
  for_each = { for i in var.letters : i => i }
}

module "shapes" {
  for_each    = { for i, k in var.shapes : (k) => i }
  source      = "zerotier/network/zerotier"
  version     = "1.0.0"
  name        = each.key
  description = each.key
  subnets     = [cidrsubnet("10.11.0.0/16", 8, each.value)]
  flow_rules  = "accept;"
}

resource "zerotier_member" "shape-letters" {
  for_each    = { for i in setproduct(var.letters, keys(module.shapes)) : j
  name        = each.key
  member_id   = zerotier_identity.letters[each.value[0]].id
  description = module.shapes[each.value[1]].description
  network_id  = module.shapes[each.value[1]].id
}
```



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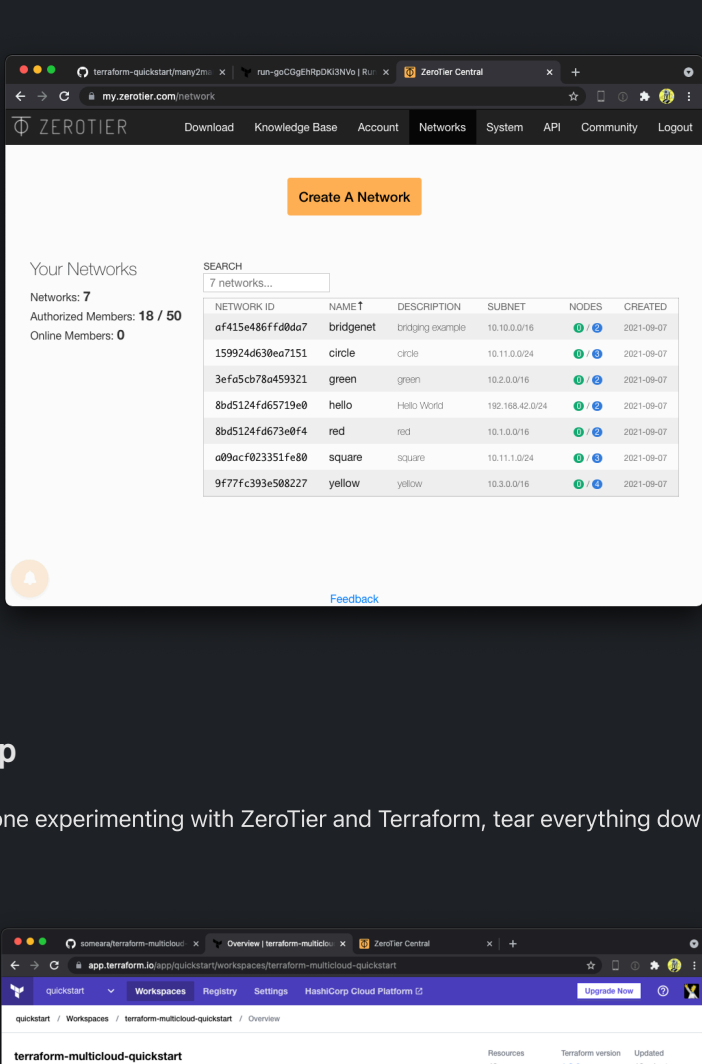
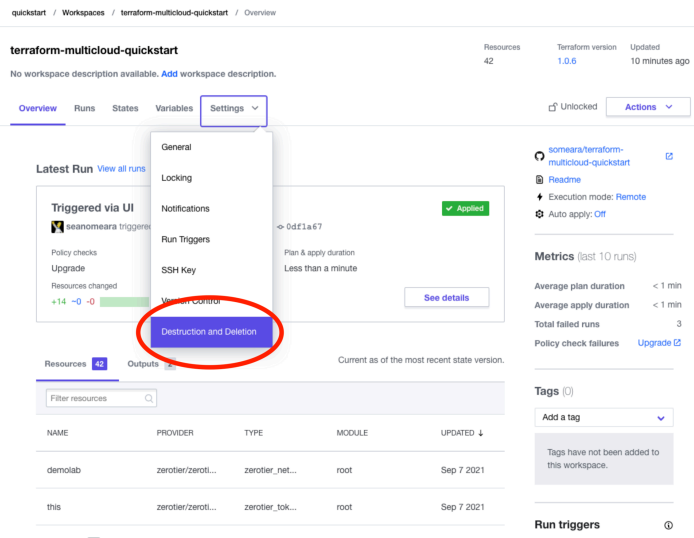
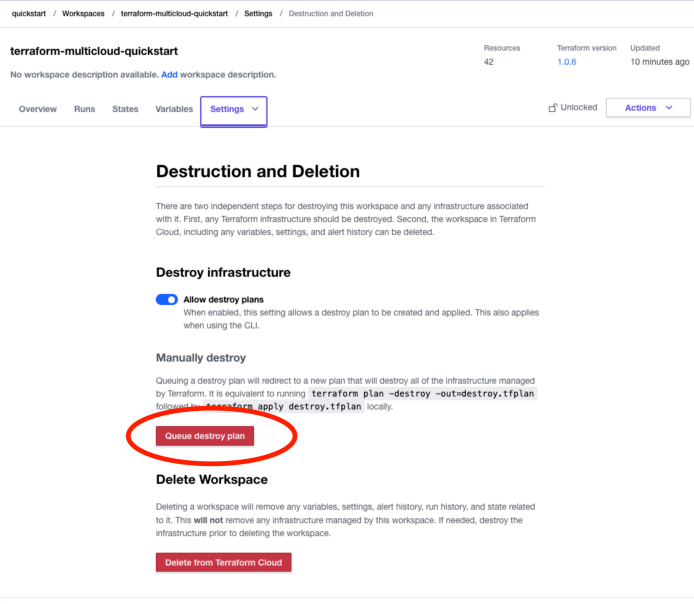


Cleaning up

When you're done experimenting with ZeroTier and Terraform, tear everything down by queueing a destroy plan.





That's all folks

If you like this tutorial, check out the ZeroTier Multicloud Terraform Quickstart next!

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